In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

In [2]:
```python
df=pd.read_csv("StudentsPerformance.csv")
```

In [3]:
```python
df.head()
```

...

In [4]:
```python
df
```

...

In [5]:
```python
df.shape
```

Out[5]:
```
(30, 6)
```

In [6]:
```python
df.dtypes
```

Out[6]:
```
Math_score                float64
Reading_score             float64
Writing_score             float64
Placement_score             int64
Club_Join_Date              int64
Placement_Offer_Count       int64
dtype: object
```

In [7]:
```python
df.isnull()
```

...

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Math_score              1
Reading_score           1
Writing_score           1
Placement_score         0
Club_Join_Date          0
Placement_Offer_Count   0
dtype: int64
```

In [9]:
```
series=pd.isnull(df["Math_score"])
df[series]
```

Out[9]:

	Math_score	Reading_score	Writing_score	Placement_score	Club_Join_Date	Placement_Offer_Count
24	NaN	81.0	66.0	81	2019	2

In [10]:
```
df.notnull()
```

...

In [11]:
```
df.describe()
```

...

In [12]:
```
df
```

...

In [13]:
```
df1=df
```

In [14]:
```
df1.fillna(0)
```

...

In [15]:
```
df2=df
```

In [16]:
```
df2["Math_score"]=df2["Math_score"].fillna(df2["Math_score"].mean())
```

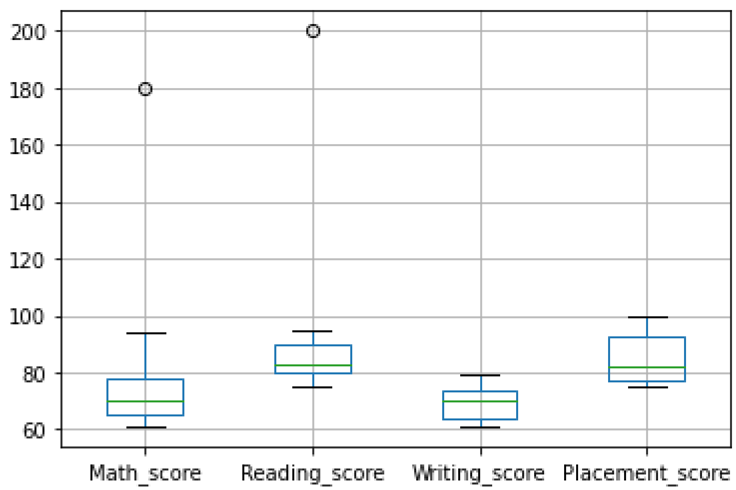
In [17]:
```
df2
```

...

In [18]:
```
col=["Math_score","Reading_score","Writing_score","Placement_score"]
```

In [19]:
```
df.boxplot(col)
```

Out[19]: <AxesSubplot:>



In [20]:
```
new_df=pd.read_csv("StudentsPerformance.csv")
new_df
```

...

In [21]:
```
print(np.where(new_df["Math_score"]>90))
```

```
(array([ 3, 17], dtype=int64),)
```

In [22]:
```
mean1=np.mean(df["Math_score"])
mean1
```
Out[22]:
```
75.10344827586208
```

//z=np.abs(stats.zscore(df['Math_score']))

In [23]:
```
z=stats.zscore(df['Math_score'])
```

In [24]:
```
z
```

...

In [25]:
```
threshold=3
```

In [26]:
```
sample_outliers = np.where(z>threshold)
sample_outliers
```
Out[26]:
```
(array([17], dtype=int64),)
```

In [27]:
```
Q1=df.Math_score.quantile(0.25)
Q1
```
Out[27]:
```
65.25
```

In [28]:
```
Q3=df.Math_score.quantile(0.75)
```

In [29]:
```
Q1
```
Out[29]:
```
65.25
```

In [30]:
```
Q3
```
Out[30]:
```
78.0
```

In [31]:
```
IQR=Q3-Q1
```

In [32]:
```
IQR
```

Out[32]:
```
12.75
```

In [33]:
```
lower_limit=Q1-1.5*IQR
upper_limit=Q3+1.5*IQR
lower_limit,upper_limit
```

Out[33]:
```
(46.125, 97.125)
```

In [34]:
```
df[(df.Math_score<lower_limit)|(df.Math_score>upper_limit)]
```

Out[34]:

	Math_score	Reading_score	Writing_score	Placement_score	Club_Join_Date	Placement_Offer_Count
17	180.0	82.0	76.0	95	2019	3

In [35]:
```
sample_outliers1=df["Math_score"][(df["Math_score"]>upper_limit)|(df["Math_score"]<lower_limit)]
```

In [36]:
```
sample_outliers1
```

Out[36]:
```
17    180.0
Name: Math_score, dtype: float64
```

In [37]:
```
index_outliers=np.where((df.Math_score<lower_limit)|(df.Math_score>upper_limit))
index_outliers
```

Out[37]:
```
(array([17], dtype=int64),)
```

In [38]:
```
redefined_df=df2
redefined_df['Math_score']=np.where((redefined_df['Math_score'] >upper_limit),mean1,redefined_df['Math_score']
```

In [39]:
```
redefined_df
```
...

In [40]:
```
col1=['Math_score']
```

In [41]:
```
redefined_df.boxplot(col1)
```
...

In [46]:
```
redefined_df.plot(kind="bar")
```
...

In [47]:
```
df_min_max_scaled=redefined_df.copy()
for column in df_min_max_scaled.columns:
    df_min_max_scaled[column]= (df_min_max_scaled[column]-df_min_max_scaled[column].min())/(df_min_max_scaled[
df_min_max_scaled
```
...

In [48]:
```
df_min_max_scaled.plot(kind="bar")
```
...

In [49]:
```
df_min_max_scaled.skew()
```
...

In [50]:
```
df_min_max_scaled.kurtosis()
```
...

In [ ]:

```
df_min_max_scaled=redefined_df.copy()
for column in df_min_max_scaled.columns:
    df_min_max_scaled[column]= (df_min_max_scaled[column]-df_min_max_scaled[column].min())/(df_min_max_scaled[
df_min_max_scaled
```

In [ ]: